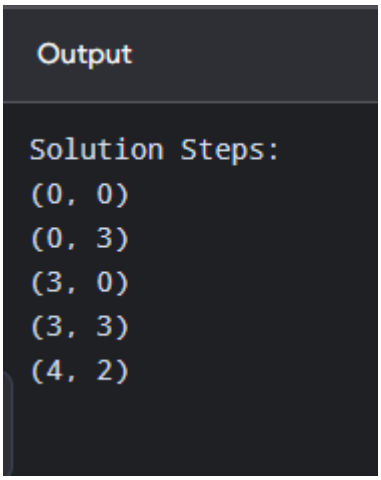
Lab experiment

Water jug problem

Coding:

```
from collections import deque

# Capacities of the jugs
A = 4
B = 3
target = 2

# BFS function
def water_jug():
    queue = deque([(0, 0)])
    visited = set()
    parent = {}

    while queue:
        x, y = queue.popleft()

        if (x, y) in visited:
            continue

        visited.add((x, y))

        if x == target or y == target:
            path = []
            state = (x, y)

            while state in parent:
                path.append(state)
                state = parent[state]

            path.append((0, 0))
            path.reverse()

            print("Solution Steps:")
            for step in path:
                print(step)

            return
```

```
        # Possible operations
        states = [
            (A, y),          # Fill Jug A
            (x, B),          # Fill Jug B
            (0, y),          # Empty Jug A
            (x, 0),          # Empty Jug B
            (x - min(x, B-y), y + min(x, B-y)),
            (x + min(y, A-x), y - min(y, A-x))
        ]

        for state in states:
            if state not in visited and state not in parent:
                parent[state] = (x, y)
                queue.append(state)

    print("No solution found")

# Main program
water_jug()
```

Output: